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Formal but Less Equal. Gender Wage Gaps in Formal and Informal Jobs in Urban Brazil

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SUMMARY

In developing countries, a large share of employees work informally and are not covered by employment protection legislation. I study how gender inequality differs across formal and informal wage-earners in urban Brazil. The raw gender wage gap is about the same on average in informal jobs (5%) as in formal jobs (7%), but I show that this difference is the result of different male and female selection processes. First, female employees have better observable characteristics than male employees, for example in terms of educational attainment. After controlling for observable characteristics, the adjusted gender wage gap is on average about 24% among formal employees and about 20% among informal employees. Second, men and women entering formal and informal jobs have different unobservable characteristics. Controlling for endogenous selection into formal *vs.* informal jobs, I find that the gender gap in wage offers is high and increases with education in formal jobs. In informal jobs, however, estimated wage offers are the same for men and women. I discuss the potential implications of these findings regarding the effect of labor market regulation on gender wage gaps.

1. Introduction

A striking characteristic of labor markets in developing countries is the existence of a large informal sector where labor regulations are non-existent. Labor regulations protect workers against several risks, for example unexpected job loss, by defining the conditions under which firms are allowed to lay off employees. Men and women do not depend in equal measure on these labor regulations since, for example, only women need to stop working around childbirth. Employees are entitled to parental leave and benefits, and job protection during leave as long as their contract has been officially registered; parental leave and benefits are almost exclusively used by women. This asymmetry affects how men and women value formal salaried jobs. It also influences the incidence of statistical discrimination against women.[1] While informal jobs have lower labor protection and labor costs, they may offer other benefits to workers such as greater flexibility. Those aspects are likely to influence female and male wages in different ways. It is thus important to understand how men and women sort across different types of jobs and to analyze how the gender wage gaps in formal and informal jobs differ. Moreover, it is useful for policymakers to understand the gender wage differences in formal and informal jobs separately since it sheds some light on how employment regulation affects women's prospects compared to men's in the labor market.

This paper focuses on employees in urban areas of Brazil and aims at examining whether significant gender gaps in earnings exist among both formal and informal wage-earners and whether the gender wage gap differs across these two groups. There are several reasons motivating the choice to focus on employees. First, it is important to analyze employees separately from other groups, such as self-employed workers, because certain determinants of gender differences in earnings apply only to employees such as employer discrimination for example. Second, there are different ways of defining informality and the criteria vary across employment statuses (Henley, Arabsheibani, & Carneiro, 2009). Looking at employees, I am able to adopt one concept of informality that depends on employers' compliance with the legislation. The formal and informal workers that I compare are similar in the sense that

[1] Discrimination implies that two individuals who are equally productive are treated unequally because they belong to different groups, defined by gender. Statistical discrimination suggests that employers have imperfect information on the behavior of their employees regarding, for example, labor market attachment as defined by the probability of quitting or duration of leave. If employment discontinuity and job protection generate higher labor costs for the firm through the re-organization of personnel as well as vacancy and replacement costs, employers have incentive to reduce expected unequally because women take *on average* more or longer periods of leave.

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they all have an employer and receive a wage. Third, focusing on employees makes it possible to compare the findings to others found in the literature, which concentrates heavily on the gender wage gaps. Lastly, employees account for the biggest share of employment. To complement the main analysis on formal and informal employees, I also provide estimates and a discussion of the gender earning gap among the self-employed.

In this paper, I do not attempt to estimate the causal effect of labor market legislation on employers' discrimination against women, but I do show that the gender gap in potential wage offers is much higher in formal salaried jobs and discuss scenarios that are consistent with this finding. To recover the gender gaps in potential wage offers I control for both observable characteristics of workers and non-random selection into formal and informal salaried jobs.[2] Failing to deal with non-random selection into formal and informal jobs is a major concern as it would lead to misleading estimates of gender wage gaps. Using the estimated gap in wage *offers*, I am then able to show what share of the total gender wage gap truly cannot be explained by workers' productive characteristics. To do so, I first use a multinomial logit model and study the sorting of men and women into different employment statuses: informal employment, formal employment, and not in paid employment. I then investigate how selection into work status affects the estimation of the gender wage gaps using the control function approach. Wage equations are estimated for formal employees and informal employees separately. In the last step, I use the Oaxaca-Blinder-Ransom decomposition to compute the gender wage gaps in both types of jobs.

The empirical analysis uses the Brazilian household survey, the Pesquisa Nacional por Amostra de Domicilios (PNAD), for the year 2015. In Brazil, all employees of the private sector have a labor card (*carteira de trabalho*) which is a booklet in which their employment biography is recorded. By signing this document, both the employer and the employee commit to abide by labor regulations and employees are entitled not only to labor rights but also to social security benefits. The PNAD provides information on whether the worker's labor card is signed by the employer so I am able to adopt a definition of informality based on employers' compliance with labor market regulations. Public sector employees are considered as formal employees.

Looking at the raw data, I find that the total gender wage gap is on average the same among formal and informal employees and increases with education in both types of jobs. Controlling for observable characteristics, I find again no significant difference between the gender wage gap among formal employees and the gender wage gap among informal employees on average.

Controlling for selection into work status, I find that the gender gap in potential wage *offers* is no longer significant among informal employees but remains significantly positive and high among formal employees. The gender gap in potential wage offers is the highest among highly educated people in formal salaried jobs but it remains insignificant across all educational levels in informal salaried jobs. These findings are consistent with a greater incidence of statistical discrimination against women and a glass ceiling effect in formal jobs. A positive gender wage gap in formal jobs is also consistent with compensating wage differentials for job protection and social security benefits when protection and benefits are more valued by women.

This paper contributes to a small strand of the literature that studies labor market outcomes for men and women when a large share of employment is informal. Pagán and Ullibarri (2000) find that in Mexico the adjusted gender wage gap is larger in unregistered firms compared to registered firms. Using data for Turkey in 1994, Tansel (2001) finds that the adjusted gender wage gap is strong and positive among workers with social security coverage but not significant among uncovered workers. Deininger, Jin, and Nagarajan (2013) focus on casual workers in India and find that the gender wage gap is particularly important for casual workers in the agricultural sector but not in non-agricultural sectors. I depart from these papers in two ways. First, in my definition of informality I focus on employers' compliance with labor regulations rather than on social security coverage or temporary work. Having information on the worker's contract rather than on firm registration enables me to consider informal wage-earners working in both registered and unregistered firms. Second, in the empirical methodology, I compare two approaches for dealing with non-random selection into multiple employment outcomes and estimate the wage function and selection bias for different education groups.

This paper is also related to the vast literature on the segmentation of the labor market and the formal wage premium. Several authors question the dualistic view of the labor market and find that some workers are actually better off choosing informal jobs (Gunther & Launov, 2012; Magnac, 1991 among others). Interview data for Brazil reported in Maloney (2004) suggest that about 30% of both male and female informal employees are voluntarily working informally. However, the motives behind this choice differ for men and women. Out of the 30% of women who preferred informal employment, about 13% cited competing household chores as the reason for this preference. Almost no men chose to work informally to balance paid employment and household responsibilities. More than 10% of men said that they did not want a formal job because they earn more in their current informal job, a statement less than 4% of women agreed with. This suggests that there might be different selection rules into informal *vs.* formal jobs for men and women. I attempt to uncover these selection rules for female and male employees and to present estimates of the gender wage gaps adjusted for the selection biases.

Finally, this paper contributes to the literature on gender wage gaps and the role of selection into employment for the assessment of gender wage gaps by adding the informality dimension (see Arabsheibani, Carneiro, & Henley, 2003; Albrecht, Bjorklund, & Vroman, 2003; Blau & Kahn, 2006; de la Rica, Dolado, & Llorens, 2008; Madalozzo, 2010; Olivetti & Petrongolo, 2008 among many others). I show that women are more positively selected than men in formal jobs but not in informal jobs. Accounting for selection has a different effect on the gender gap in wage offers in formal and informal jobs.

The remainder of the paper is organized as follows. I start by discussing the impact of informality on gender employment and wage inequality while reviewing the related literature. Section 3 sets up the empirical model. In Section 4 I first describe the data and provide descriptive statistics on gender inequality among wage-earners in the urban labor markets of Brazil. I then discuss the results, looking at the selection into potential work-statuses before moving on to the comparison of the gender earnings gaps in formal salaried jobs, informal salaried jobs, and self-employment. The last section concludes.

2. Why should informality matter?

Before considering the empirical evidence, I first ask why the gender wage gap should differ across formal and informal jobs.

Individual characteristics across different types of jobs. According to the dualistic view of the labor market, the informal sector is characterized by lower wages. Empirical evidence confirms that

[2] Non-random selection into jobs means that, even after controlling for observable characteristics, formal employees are systematically different from informal employees in a way that affects their wages. This is also referred to as "selection based on unobservables".

formal jobs offer on average higher wages than informal jobs.[3] Comparisons of raw average wage gaps are informative about the *accepted* wage offers but conceal heterogeneity in workers' observable and unobservable attributes. Empirical papers show that the formal–informal wage difference varies with workers' skill levels. Studying the urban labor market in Mexico, Gong and Van Soest (2002) find a significant wage premium in formal jobs for educated men but not for men with a low level of education, who earn more on average in informal jobs. For women, the differences between formal and informal wages are small. It is thus important to take into account men's and women's productive characteristics, such as education levels, and compare individuals with similar observable productivity within job types. Moreover, men and women may also sort themselves into the different types of jobs because of unobservable characteristics that may correlate with wages. To infer how informality affects gender gaps in potential wage offers, one needs to account for this endogenous sorting of men and women into formal and informal jobs.

Compensating wage differentials. What might the explanation be for different gender wage gaps in formal and informal jobs once we have controlled for individual characteristics? From the labor supply side, if individuals have different preferences for the type of job, the theory of compensating wage differentials can provide an explanation for differences in the gender wage gaps across formal and in informal jobs. Formal jobs offer non-wage benefits that are not available in informal jobs such as job severance payments, maternity leave and benefits, and unemployment benefits. In a frictionless market, workers with identical productivity should earn a higher wage in informal jobs to compensate for the absence of these benefits. If women value job protection more than men, for reasons linked to motherhood for example, women should be willing to accept lower wages compared to men in the formal sector but not in the informal sector. This would lead to a gender wage gap among formal employees only. However, if women value the flexibility of informal jobs more than men to better combine paid work and household responsibilities, one should also observe a gender wage gap among informal employees. There are reasons to value the amenities of both sectors and workers' preferences for formal or informal employment hinge on the balance of the advantages and disadvantages of both statuses. Gender differences in preferences are not a priori clear cut, which makes it difficult to draw theoretical predictions on the overall effect of preferences on gender wage gaps across sectors.

Labor regulation and statistical discrimination. From the labor demand side, job offers stem from both registered and unregistered firms. While firms operating informally cannot offer legal contracts to their employees, firms operating formally might decide to hire workers formally or informally. Why would employers set different wages for a man and a woman with similar observable characteristics and employed under the same type of contract? Employers weigh up the costs and benefits of labor contract registration for both men and women and determine their hiring decisions and wage setting rules accordingly. In case of imperfect information on the behavior of their employees, regarding for example labor market attachment as defined by the probability of quitting or duration of leave, employers have incentive to use information on typical behavior of the demographic group to which the employee belongs. In other words, employers have an incentive to treat a man and a woman differently because women have *on average* lower labor market attachment, even if a particular man and a particular woman are equally productive on the job. Employers may expect a higher quit rate among women because they are more likely to take permanent or temporary leave to have children, for example. Lazear and Rosen (1990) provide a theoretical explanation where greater domestic responsibilities generate higher female quit rates and lower female wages due to statistical discrimination.[4] Employment discontinuity generates higher labor costs for the firm through vacancy and replacement costs; it can also generate forgone profits if no one can replace the employee on leave or if the time out of the job causes a loss of (general or specific) skills. Employers may want to compensate for the higher female quit rate by paying women lower wages. This argument applies especially to formal jobs where employers abide by labor regulations such as job protection during maternity leave. In other words, statistical discrimination against women should lower female wages in formal jobs where employers' costs associated with their employees' job discontinuity are higher. The impact of labor regulation on gender wage gaps has been in part addressed in papers that investigate the effects of family policies on gender outcomes. Ruhm (1998) exploits variation in maternity leave coverage across European countries between 1969 and 1993. He concludes that more generous leave policies are associated with increased employment but with lower wages for women in comparison to men. Gruber (1994) uses variation in the timing of a reform in health insurance laws across states of the US. The reform increased employers' relative costs of insuring women of childbearing age and his findings indicate that much of these costs were shifted to female wages. Using data for several developed economies, Mandel and Semyonov (2005) find that longer maternity leave corresponds to a wider gender pay gap.

In the case of Brazil, women in formal jobs are entitled to 120 days of paid maternity leave while men are entitled to up to 5 days of paid paternity leave. Moreover, job protection regulations prevent employers from making women redundant in the 5 months following childbirth. In informal jobs, women do not benefit from job protection nor maternity leave benefits. In this context, the expectation is that the gender wage gap is bigger in formal jobs because of statistical discrimination and/or unequal compensating wage differentials. I am not aware, however, of any study that has been able to exploit a labor or family policy reform in a context of a dual labor market where a significant share of the workforce is not covered by mandated benefits. In the absence of exogenous change in regulation, this paper does not provide any direct evidence on the effect of employment protection legislation on the wages of female employees compared to male employees. Instead I compare the gender wage gaps across salaried jobs with and without mandated benefits and employment protection, and discuss how the findings are consistent with the arguments discussed above. The next section presents the econometric methods used to explore whether the gender wage gap is higher in formal jobs.

3. The econometric model

I compare the gender wage gaps among formal and informal employees and investigate how selection shapes the gender wage gaps in these two different groups. I first compute the raw wage gaps and the wage gaps adjusted for observable characteristics. Next, I compute the wage gaps controlling for both observable characteristics and the selection into the different labor statuses. By comparing the raw and the adjusted wage gaps, one can see how gender differences in observable characteristics and in selection into formal and informal jobs shape gender wage inequality.

[3] See Magnac (1991), who analyzes female wages in Colombia, Gasparini and Tornarolli (2009), who study different Latin American countries, and Almeida and Carneiro (2012), who find that the formal raw wage premium is positive in Brazil and decreases with regulation enforcement.

[4] See also Bertrand, Goldin, and Katz (2010).

(a) The raw and the adjusted wage gaps

The raw wage gap in sector j is estimated from an equation where $\ln w_{ij}$ the hourly log wage is regressed on a constant and a female dummy only:

$$\ln w_{ij} = \beta_0 + \alpha_j F_{ij} + u_{ij}, \tag{1}$$

where $F_{ij} = 1$ if employee i working in sector j is a woman. The raw wage gap in sector j is: $E(\ln w_j|female) - E(\ln w_j|male) = \widehat{\alpha}_j$.

To compute the adjusted wage gap, I use a version of the wage gap decomposition developed by Oaxaca (1973) and Blinder (1973) that avoids important methodological problems discussed in Oaxaca and Ransom (1994, 1999). The decomposition methodology that follows has been presented in Fortin (2008) and is not sensitive to the choice of the reference wage structure. The reference wage structure is taken from the estimation of a common wage regression on the pooled sample of both men and women where the male advantage equals the female disadvantage with respect to the reference. I estimate three equations: (i) a pooled wage equation with subscript p including gender dummies F_i and M_i and an identification restriction, (ii) a wage equation for men with subscript m, and (iii) an equation for women with subscript f. Each equation is estimated separately for formal and informal jobs denoted with the subscript $j = \{\inf, \text{for}\}$.

$$\ln w_{ipj} = \beta_{0pj} + \alpha_{pfj} F_i + \alpha_{pmj} M_i + \mathbf{X_i}\beta_{pj} + u_{ij} \quad \text{with } \alpha_{pfj} = -\alpha_{pmj}, \tag{2a}$$

$$\ln w_{ifj} = \beta_{0fj} + \mathbf{X_i}\beta_{fj} + u_{ifj}, \tag{2b}$$

$$\ln w_{imj} = \beta_{0mj} + \mathbf{X_i}\beta_{mj} + u_{imj}, \tag{2c}$$

where X is a set of control variables that includes the number of years of education, the age and the age squared, the tenure in the current job and its square, whether the person is black, whether the person lives in an urban area, and indicators for regions and sectors of activity. To capture demand side effects, I use regional unemployment rates that characterize the state of the local labor market. I construct the regional unemployment for different education groups in order to identify the impact of lower labor demand even when controlling for regional dummies. The assumption is that labor markets are, at least to some extent, skill-specific. Even if workers accept a job for which they are overqualified, the unemployment rate among people of the same (generally defined) skill level will impact their decision to participate, their job-finding rate, and their wages.

The zero conditional mean assumption $E(u_m|x_m) = E(u_f|x_f) = 0$ ensures that the error is uncorrelated with the regressors so that the OLS estimates are unbiased. The zero conditional mean assumption also ensures that the total average wage gap can be exactly decomposed into terms based on observables and their returns. For the wage decomposition to be exact though, only a weaker ignorability assumption is sufficient: the distribution of u given X must be the same for the two groups. In other words, the decomposition allows for selection on unobservables as long as they yield identical selection biases for both men and women.[5] Under the ignorability assumption, the total wage gap in each type of job j can be decomposed into three terms:

$$\overline{\ln W}_{mj} - \overline{\ln W}_{fj} = (\overline{\mathbf{X}}'_m - \overline{\mathbf{X}}'_f)\widehat{\boldsymbol{\beta}_{pj}} + \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}_{pj}}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}_{pj}} - \widehat{\boldsymbol{\beta}_{fj}}).$$

The first term on the right-hand side accounts for gender differences in characteristics; it is the endowment term. The rest accounts for gender differences in the prices associated with given characteristics; it is also called the coefficient term and is here decomposed into the male advantage with respect to the reference prices and the female disadvantage with respect to the reference prices. The adjusted wage gap is the sum of the male advantage and the female disadvantage in the treatment of the characteristics:

$$\overline{WG}_j = \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}_{pj}}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}_{pj}} - \widehat{\boldsymbol{\beta}_{fj}}). \tag{3}$$

However, several observations indicate that selection into employment may follow different processes for men and women. First, women have a much lower labor market participation rate than men. Second, among individuals participating in the labor market, the female unemployment rate is higher than the male unemployment rate and the informality rate is higher among female employees compared to male employees. If the ignorability assumption does not hold, men employed in a given type of job have different unobservables than women employed in the same type of job. To check for and eliminate selection biases I adopt a control function approach that is presented in the next sub-section.

(b) Treatment for selection into multiple employment statuses

I present here how I construct the control function used in the estimation of the gender wage gaps among employees. The sample excludes the self-employed, employers, and unpaid family workers. To model individuals' choices (constrained or unconstrained), I consider here a one-step process (Figure 1). I use a multinomial model to estimate the probability of being in three mutually exclusive outcomes denoted j: not in paid employment $(Y_i = 1)$, informal wage-employment $(Y_i = 2)$, and formal wage-employment $(Y_i = 3)$. In this setting, individuals can have different probabilities of being in a given work status depending on their characteristics and preferences, as well as on demand constraints and employers' behavior that may cause job rationing and segregation.

The latent value (or utility) associated with being in state j is denoted V_{ij}. State j is observed $Y_i = j$ if the value associated with this state is higher than the value of the other states, or in other words, if status j is the best available option for individual i:

$$Y_i = j \text{ if } V_{ij} > \max_{k \neq j}(V_{ik}).$$

I assume that the utility associated with work status j follows a linear function: $V_{ij} = \mathbf{X}_{1i}\boldsymbol{\lambda}_j + \mathbf{Z}_i\boldsymbol{\alpha}_j + \mu_{ij}$, for $j = 1, \ldots, 3$. The full model of selection and wage determination can be written as follows

$$\ln w_{ij} = \mathbf{X}_{1ij}\boldsymbol{\beta}_{1j} + \mathbf{X}_{2ij}\boldsymbol{\beta}_{2j} + u_{ij}, \quad \text{if } V_{ij} > \max_{k \neq j}(V_{ik}) \quad \text{for } j = 2, 3, \tag{4}$$

$$V_{ij} = \mathbf{X}_{1i}\boldsymbol{\lambda}_j + \mathbf{Z}_i\boldsymbol{\alpha}_j + \mu_{ij}, \quad j = 1, \ldots, 3, \tag{5}$$

where individual i earns a wage w_{ij} if she is a formal worker $j = 2$ or an informal worker $j = 3$ and j is the observed outcome if the value associated with state j is the highest. The vector $\mathbf{X}_{1i}$ includes productive characteristics of individual i, namely years of education, age and age squared, whether the person is black, a macroeconomic demand side variable to capture rationing: regional unemployment rate by education group, and regional dummies. The vector $\mathbf{X}_{2ij}$ includes wage determinants that are only observable if individual i is in paid employment: tenure in the current job, tenure squared, and indicators for the sector of activity. In the selection equation, $\mathbf{X}_{1i}$ includes wage determinants as they influence the choice of the work status. The vector $\mathbf{Z}$ additionally includes variables that do not directly affect wages but are relevant to the work status determination in the selection equation. I discuss the set of excluded variables in detail below.

[5] See Fortin, Lemieux, and Firpo (2011) for a discussion of the assumptions required for identification in wage decompositions.

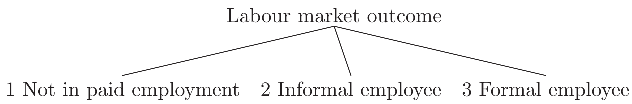

Figure 1. Work status, simultaneous decision.

A selection bias arises in the wage Eqn. (4) if the unobserved characteristics that influence wages u_{ij} are correlated with the unobserved determinants of the selection process μ_{ij}. In other words, the estimates of the wage equation are biased if the individuals who are more likely to find a formal (informal) job have characteristics -unobserved in the data- that affect wages in formal (informal) jobs. In this case, observed wages in formal (informal) jobs differ from the potential wage offers (that would be) made to all men and women if they were to try and get a formal (informal) job. To understand how one can deal with selection bias, it is useful to use mathematical notations again and express the conditional expectation of the wage as: $E(w_j|Y=j,\mathbf{X}_1,\mathbf{Z})=\mathbf{X}_{1j}\boldsymbol{\beta}_{1j}+\mathbf{X}_{2j}\boldsymbol{\beta}_{2j}+E(u_j|Y=j,\mathbf{X}_1,\mathbf{Z})$. If selection into jobs is not random the last term is not equal to zero and omitting it can lead to biased estimates. In other words, non-random selection into types of jobs generates an omitted variable bias. The approach to dealing with non-random selection consists precisely of introducing a correction term equal to the conditional mean of the error term $E(u_j|Y=j,\mathbf{X}_1,\mathbf{Z})$ in the wage equation. The correction term is called the control function and is denoted here $h(.)$. Several methods are available to compute the control function; they differ in their assumptions on the covariances between the error term of the wage equation and the error terms of the outcome equations. I use and compare two methods, Lee's (1983) approach and the Dubin–McFadden approach, that are described in the Appendix.

The following wage equations corrected for selection are then estimated by least squares:

$$\ln w_{ipj}=\alpha_{pfj}F_i+\alpha_{pmj}M_i+\mathbf{X}_i\beta_{pj}+\boldsymbol{\theta}_{pj}h_{pj}(P_1,\ldots,P_3)+u_{ij} \quad \text{with } \alpha_{pfj}=-\alpha_{pmj}, \tag{6a}$$

$$\log w_{ifj}=\mathbf{X}_{ij}\boldsymbol{\beta}_{fj}+\boldsymbol{\theta}_{fj}h_{fj}(P_1,\ldots,P_3)+\epsilon_{ifj}, \tag{6b}$$

$$\log w_{imj}=\mathbf{X}_{ij}\boldsymbol{\beta}_{mj}+\boldsymbol{\theta}_{mj}h_{mj}(P_1,\ldots,P_3)+\epsilon_{imj}, \tag{6c}$$

where $\boldsymbol{\theta}_j h_j(P_1,\ldots,P_3)=E(u_j|X,Y=j)$ and depends on the model assumptions. The estimation of Eqn. (6) allows us to recover ρ_j, the correlation between u_j and μ_j, when Lee's model is adopted, and the correlation between u_j and all the μ_k for $k=\{1\ldots j\ldots 3\}$ if the Dubin–McFadden approach is used.

The total decomposition with the additional term that captures the difference in average selection bias is expressed as follows:

$$\overline{\ln W}_{mj}-\overline{\ln W}_{fj}=(\overline{\mathbf{X}}'_m-\overline{\mathbf{X}}'_f)\widehat{\boldsymbol{\beta}_{pj}}+\overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj}-\widehat{\boldsymbol{\beta}}_{pj})+\overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}}_{pj}-\widehat{\boldsymbol{\beta}}_{fj}) + \boldsymbol{\theta}_{mj}\mathbf{h}_{mj}(P_1,\ldots,P_3)-\boldsymbol{\theta}_{fj}\mathbf{h}_{fj}(P_1,\ldots,P_3).$$

where the selection term provides a measure of the difference between the observed wage gap and the gap in wage offers.[6] The wage gap due to different returns to observable characteristics in sector j is:

$$\overline{WG}_{Sj}=\overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj}-\widehat{\boldsymbol{\beta}}_{pj})+\overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}}_{pj}-\widehat{\boldsymbol{\beta}}_{fj}). \tag{7}$$

The adjusted wage gap in Eqn. (7) differs from the one in (3). First, the coefficients are now consistently estimated following the treatment for selection. Second, instead of explaining part of the total observed wage gap, the difference in returns now explains the gap in wage offers $\overline{\ln W}_{mj}-\overline{\ln W}_{fj}-(\theta_{mj}h_{mj}(P_1,\ldots,P_3)-\theta_{fj}h_{fj}(P_1,\ldots,P_3))$.

Eqn. (6) is also estimated for various education groups separately to explore how the selection rules and the gender gaps in wage offers differ across groups.

(c). Identification

To identify the effect of selection and purge the wage estimates from selection bias without relying on the functional forms, I make exclusion restrictions. I assume that the following variables determine the potential work status but do not directly affect wages: the share of household members holding formal jobs, the presence of children, marital status, an indicator for single mothers, and the share of household members who need care (children under the age of 10 and elderly people over 85 years). The latter variable may be more appropriate than just the number of children given that, first, older children may take care of younger children, and second, older family members may live in the same household and receive care. It may be argued that children, or in general relatives who need care, can affect the productivity of women at work (through fatigue and lower availability) and thus may not be an appropriate excluded variable. For this reason, I check the sensitivity of the results to the choice of exclusion restrictions and present results both with and without children as an excluded variable. The share of household members holding a formal job can determine the probability of being in formal *vs.* informal employment through a network effect. Arguably, an individual with household members who hold a formal job will receive more information about job openings in the formal sector, how and where to search for formal jobs and/or might be recommended for a job. This networking effect may also be thought of as a cultural norm or social capital channel.[7] Moreover, once individuals have found a job, the share of household members holding formal jobs has a priori no effect on wages.

The empirical approach will hence consist of computing the formal (informal) gender wage gaps, firstly controlling for observable characteristics only, and secondly controlling additionally for endogenous selection into formal (informal) employment. The latter specification requires estimating the probability of being in each outcome in a first step using a multinomial logit.

4. Empirical evidence

(a) Data for the Brazilian urban labor market

Individual information is taken from the 2015 Brazilian household survey, the Pesquisa Nacional por Amostras de Domicilio (PNAD).[8] This paper focuses on urban areas.[9] For 2015, the PNAD provides information on individuals from roughly 117,900 urban households. Around 199,600 working-age people (18–65) were interviewed, of whom 52% were women. Both the public and private sectors are represented. I exclude the few individuals working in the military as it is almost completely male-dominated. I focus on gender differences among formal and informal *wage-earners*, including domestic workers but excluding the self-employed, employers, unpaid, and family workers. This way, the formal and informal workers that I compare are similar in the sense that they all have an employer and receive a wage. Moreover, the survey provides direct

[6] This approach to selection has been applied to gender wage gap decomposition by Wright and Ermisch (1991) for the U.K., Ogloblin (1999) for Russia, Appleton, Hoddinott, and Krishnan (1999) for three African countries and Jung (2017) for Korea.

[7] Rauch (1991) mentions this as the "inheritance" of formal and informal sector status due to "connections" made by relatives working in the formal sector, and stresses that it reenforces inequality persistence across families.

[8] The same analysis for 2009 can be found in Ben Yahmed (2016).

[9] In 2015, more than 85% of the Brazilian population lived in urban areas.

and reliable information that enables me to classify employees as either formal or informal wage-earners. Individuals are asked if their labor card is signed by their employer; if it is not, they are not registered and are not entitled to any labor rights or benefits. The labor card is used in the private sector; workers in the public sector have other types of contracts and are considered as formal employees in this study. Employees account for the biggest share of employment. In 2015, 64% of men and 68% of women in the labor force were employees. Self-employment accounted for 22% and 12% of the male and female labor force respectively, and 5% of men and 2% of women were employers.[10]

Table 1 gives the demographic, household, and educational characteristics of men and women holding formal and informal jobs and those not in paid employment. Informal employees are on average younger than formal employees. Women who hold informal jobs are more often the head of the household, are less likely to live with a partner compared to women in formal jobs and are more likely to have young children. But these characteristics are especially likely among women who are not in paid employment: 81% of them live with a partner, and 45% of them have young children. In contrast, men who are not in paid employment are unlikely to have children. Wage-earners in formal employment tend to live in households with a higher share of formal wage-earners, which seems to point to the network argument. The share of household members in formal employment is 11 percentage points higher among formal female workers compared to informal female workers and 8 percentage points higher compared to women not in paid employment. Women tend to live in households with a higher share of formal wage-earners compared to men; this differential can be explained by the higher male participation rate and lower male informality rate compared to the corresponding female rates, a difference that is discussed below. In terms of education, women have higher educational attainment than men in all three employment statuses. The difference is the strongest among formal employees and points toward a more positive selection of women into formal jobs.

Turning to job characteristics, as documented in the literature on informality, wages are higher on average in formal jobs. Women earn lower hourly wages than men in both types of jobs. They also work fewer hours than men, especially in informal salaried jobs. Among formal wage-earners, women are more likely to work in the public sector (29% of women compared to 15% of men). There are no major differences across gender or sector in the average tenure in the current job, which is somewhat surprising since one would expect higher job turnover and thus lower tenure in informal jobs.

Table 2 gives further evidence of positive selection of women into salaried employment. The participation rate is significantly lower among women, especially among people with primary education or less, with only 45% of women participating in the labor market compared to 78% of men, which corresponds to a gap of 33 percentage points. The participation rate increases with education and more rapidly for women. Among people with tertiary education, the participation gap is 10 percentage points. Columns 2 and 3 show that the female unemployment rate is higher than the male unemployment rate in all education groups; the difference is the smallest for people with a high level of education.

The last two columns report the informality share among employees which measures the share of wage-earners without a registered labor contract. The informality rate as well as the gender gap in the informality rate decrease with education. Among wage-earners with primary education or less, 49% of women are employed without a formal contract, a figure that is 16 percentage points higher than for men. Among wage-earners with tertiary education, however, the informality rate is about the same for men and women at 10%.

I now turn to the distribution of female and male employment across sectors and to the informality rate within sectors. As reported in Table 3, 90% of female employees work in the service sector where the informality rate for women is 24%. Only 64% of male employees work in this sector and they have a lower informality rate at 15%. The highest informality rate is in agriculture but this sector employs a very small share of wage-earners in urban areas.

Table 4 gives the distribution of female and male employment across occupations and the informality rate within each occupation. The informality rate is the lowest among high-skill occupations such as managers and officials and professionals. Excluding agricultural workers, the informality rate is the highest for women in the service and sales occupation where 40% of them are informally employed. This occupation employs almost the majority of female employees. In contrast, only 24% of men work as service and sales workers and 18% of them are informally employed.

(b) Selection into multiple potential employment statuses

I start the empirical analysis by estimating the multinomial logit equation to understand the impact of supply-side and demand-side variables on the probability of being a formal employee, an informal employee, or not in paid employment. I estimate the multinomial logit model for men and women separately. The marginal effects are reported in Tables 5 and 6. The tables provide an estimate of the effect of a marginal change in each continuous variable, or a discrete change for indicators, for an individual with average characteristics in the male sample and in the female sample.

The probability of working as a formal employee increases with age for both men and women. An additional year of education reduces the probability of being out of the labor force and increases the chances of being formally employed, and these effects are stronger for women.[11]

Other variables such as the family structure have opposite effects for men and women. Women with young children and living with a partner are less likely to be in paid employment while it is the opposite for men. This does not hold for single mothers, however, who are more likely to hold a formal or informal job compared to women without children (the effect of being a single mother overcompensates the effect of having children on each outcome probability). Unlike women with young children living with a partner, men with young children are more likely to be employed, most likely in a formal job. These results are consistent with the traditional division of roles within the household.

People in a household with a higher share of formal workers are more likely to hold a formal jobs and less likely to hold an informal job. The effect is stronger for men. This finding points toward the network effect discussed in the previous section.

Looking at demand-side factors, a higher regional unemployment rate, calculated for each skill and gender group, increases the probability of non-participation and of holding an informal job for women but not for men. It reduces the probability of finding a formal job for women, whereas the opposite holds for men, although the effect is not significant. This may reveal an insurance effect: as it becomes tougher to find a job, men tend to search more

[10] The remaining work statuses in the labor market are unemployment (concerning 8% of men and 12 % of women) and unpaid employment (accounting for 1% of men and 2% of women in the labor force).

[11] Statements on gender differences are based on differences in point estimates. I do not formally test for the difference between the male and the female coefficients here. However, the number of observations is large and all coefficients are precisely estimated so the gender differences are likely to be significant.

Table 1
Descriptive statistics by gender for Brazilian urban areas

	Formal				Informal				Not in paid employment			
	(1) Men		(2) Women		(3) Men		(4) Women		(5) Men		(6) Women	
Demographics												
Age (mean)	35.2	(10.1)	35.8	(10.0)	32.2	(10.5)	35.9	(10.6)	31.0	(11.3)	35.29	(11.6)
Head of household	0.55	(0.50)	0.28	(0.45)	0.44	(0.50)	0.31	(0.46)	0.26	(0.44)	0.24	(0.43)
Living with a partner	0.80	(0.40)	0.75	(0.44)	0.74	(0.44)	0.73	(0.45)	0.67	(0.47)	0.81	(0.39)
Children under 14	0.40	(0.49)	0.36	(0.48)	0.39	(0.49)	0.39	(0.49)	0.26	(0.44)	0.45	(0.50)
Household composition												
Share of the household members with a formal job[a]	0.27	(0.33)	0.34	(0.34)	0.16	(0.27)	0.23	(0.31)	0.23	(0.30)	0.26	(0.30)
Share of household members who need care	0.12	(0.17)	0.11	(0.16)	0.13	(0.17)	0.12	(0.17)	0.09	(0.15)	0.16	(0.18)
Education												
Illiterate	0.02	(0.15)	0.01	(0.11)	0.07	(0.25)	0.05	(0.22)	0.09	(0.29)	0.05	(0.23)
Years of schooling(mean)	10.13	(3.53)	11.47	(3.31)	7.80	(3.9)	8.5	(3.8)	8.2	(4.1)	8.8	(3.8)
Job-related variables												
Hourly wage	20.6	(115.5)	19.0	(87.3)	9.4	(25.2)	8.7	(21.2)				
Hours of work	42.2	(8.5)	39.4	(9.0)	40.7	(11.7)	32.7	(14.2)				
Full time	0.93	(0.25)	0.84	(0.36)	0.83	(0.38)	0.55	(0.50)				
Several jobs	0.03	(0.17)	0.04	(0.19)	0.02	(0.15)	0.03	(0.16)				
Public sector	0.15	(0.36)	0.29	(0.45)								
Tenure (mean number of years)	2.96	(3.21)	2.93	(3.22)	2.68	(2.86)	2.72	(2.98)				
Nightwork	0.04	(0.19)	0.01	(0.11)	0.02	(0.14)	0.01	(0.09)				
N	26,059		20,644		5,994		5,287		8,443		21,032	

Source: Author's calculations based on the PNAD 2015, IBGE, Brazil. The columns give the shares among male formal wage-earners (1), female formal wage-earners (2), male informal wage-earners (3), female informal wage-earners (4), males not in paid employment (5), and females not in paid employment (6) in urban labor markets. Standard deviations in parentheses.

[a] The share of working-age household members holding formal jobs excludes the respondent. I will use this variable to explain the sorting of individuals across job types.

Table 2
Descriptive statistics by education groups and gender

Education	Participation rate		Unemployment rate		Informality rate among employees	
	Men	Women	Men	Women	Men	Women
Total	86	64	8	12	18	23
Primary or less	78	45	7	9	33	49
Secondary	88	66	9	15	18	25
Tertiary	87	77	7	8	10	9

Source: Author's calculations based on the PNAD, 2015, IBGE, Brazil. Individuals living in urban areas only. All numbers are in percentage.

Table 3
Employment shares and informality rate by gender and sectors in urban areas

Sector	Employment share		Informality rate	
	Women	Men	Women	Men
Agriculture	1	5	43	54
Construction	1	13	14	38
Manufacturing	8	18	12	10
Services	90	64	24	15

Source: Author's calculations based on the PNAD, 2015, IBGE, Brazil. Urban areas only. All numbers are in percentage. The employment share in sector *j* is equal to the level of employment in sector *j* divided by total employment. The informality rate in sector *j* is equal to the level of informal wage-employment in sector *j* divided by total wage employment in sector *j*.

intensively for formal jobs that are more secure and provide unemployment benefits in case of redundancy.

(c) Wages in formal and informal jobs across gender

Table 7 presents the estimates of the female wage equations and Table 8 presents the estimates of the male wage equations. In each table, columns (1) and (2) give the OLS estimates of a standard wage equation and columns (3) to (6) give the estimates of a wage equation augmented with the control function.[12]

The return on education is stronger in formal jobs for both women (Table 7) and men (Table 8). This pattern is robust to the introduction of the selection control function. Wages also increase with age and there is no evidence of a concave effect of age on wages. Experience at the current firm increases women's and men's wages in formal jobs. In informal jobs, however, firm-specific experience has a negative effect that is significant for men only. As I control for age, this result does not mean that actual wages decline over time for men employed in informal jobs. Negative returns on firm-specific experience in informal jobs mean that workers who keep on working informally for the same employer have lower wages compared to workers who have changed job more recently. The negative returns of tenure with the same employer may be due to low job mobility along with monopsony power of employers. The regional unemployment rate negatively affects wages, especially in formal jobs; this effect is also robust to selection treatment.

[12] For this step, I use the Stata command selmlog that implements the methods described and evaluated in Bourguignon, Fournier, and Gurgand (2007). See http://www.parisschoolofeconomics.com/gurgand-marc/selmlog/selmlog13.html.

Table 4
Employment shares and informality rate by gender and occupations

Occupation	Employment share		Informality rate	
	Women	Men	Women	Men
Managers and Officials	3	4	7	8
Professionals	16	8	6	10
Technicians and associate professionals	8	8	7	12
Clerical support workers	18	9	7	7
Service and sales workers	47	26	40	18
Skilled agricultural and forestry workers	1	4	46	55
Craft and related trades workers	4	32	18	24
Operators, and assemblers	1	7	9	18

Source: Author's calculations based on the PNAD, 2015, IBGE, Brazil. Urban areas only. All numbers are in percentage. The employment share in occupation *j* is equal to the level of employment in occupation *j* divided by total employment. The informality rate in occupation *j* is equal to the level of informal wage employment in occupation *j* divided by total wage employment in occupation *j*.

Table 5
Labor market status, marginal effects for women

Women	Not in paid employment	Informal employee	Formal employee
Age	−0.041***	0.007***	0.034***
	(0.001)	(0.001)	(0.001)
Age^2	0.001***	−0.000***	−0.000***
	(0.000)	(0.000)	(0.000)
Years of education	−0.028***	−0.010***	0.038***
	(0.001)	(0.000)	(0.002)
Non-white	−0.014*	0.016***	−0.001
	(0.006)	(0.003)	(0.006)
Having children	0.011*	−0.007	−0.004
	(0.005)	(0.004)	(0.005)
. . .under 14	0.017***	−0.009*	−0.008
	(0.004)	(0.004)	(0.007)
Head of household	−0.051***	0.027***	0.024**
	(0.008)	(0.004)	(0.008)
Living with a partner	0.079***	−0.020***	−0.060***
	(0.007)	(0.004)	(0.007)
Single mother	−0.056***	0.032***	0.025**
	(0.007)	(0.007)	(0.009)
Share of formal workers	−0.045***	−0.051***	0.096***
in the household[a]	(0.009)	(0.005)	(0.007)
Share of individuals	0.237***	−0.046***	−0.192***
who need care	(0.016)	(0.004)	(0.017)
Born in the municipality	−0.087***	0.001	0.087***
	(0.007)	(0.007)	(0.010)
Unemployment rate	0.882***	0.383***	−1.265***
(regional, by education group)	(0.087)	(0.078)	(0.097)

Notes: Marginal effects, standard errors in parenthesis. The marginal effects of each explanatory variables on the probability of being in each of the three different outcomes are computed based on a multinomial logit estimation.
[a] The share of working-age household members holding formal jobs excludes the respondent. Number of observations 75,408.

Table 6
Labor market status, marginal effects for men

Men	Not in paid employment	Informal employee	Formal employee
Age	−0.027***	−0.007***	0.034***
	(0.001)	(0.001)	(0.001)
Age^2	0.000***	0.000***	−0.000***
	(0.000)	(0.000)	(0.000)
Years of education	−0.015***	−0.013***	0.028***
	(0.001)	(0.001)	(0.001)
Non-white	−0.004	−0.000	0.005
	(0.004)	(0.004)	(0.006)
Having children	0.046***	−0.014***	−0.032***
	(0.003)	(0.004)	(0.004)
. . .under 14	−0.057***	0.008*	0.049***
	(0.006)	(0.004)	(0.007)
Head of household	−0.088***	0.006	0.082***
	(0.005)	(0.005)	(0.007)
Living with a partner	−0.065***	−0.008*	0.073***
	(0.003)	(0.003)	(0.003)
Share of formal workers	−0.039***	−0.108***	0.146***
in the household[a]	(0.007)	(0.009)	(0.010)
Share of individuals	−0.039*	0.002	0.037*
who need care	(0.016)	(0.013)	(0.017)
Born in the municipality	−0.038***	−0.006	0.044***
	(0.009)	(0.009)	(0.010)
Unemployment rate	−0.069	−0.074	0.143
(regional, by education group)	(0.129)	(0.087)	(0.143)

Notes: Marginal effects, standard errors in parenthesis. The marginal effects of each explanatory variables on the probability of being in each of the three different outcomes are computed based on a multinomial logit estimation.
[a] The share of working-age household members holding formal jobs excludes the respondent. Number of observations: 57,218.

In columns (3) to (6) of Tables 7 and 8, the control function is included as an additional regressor. The selection bias is significant in both formal and informal jobs and for both men and women. The correlation coefficient gives us the direction of the average selection rules for men and for women. Note that when Lee's approach is adopted, a negative $\sigma_j\rho_j$ implies here a positive selection bias as $\sigma_j\rho_j\left(-\frac{\phi(\Phi^{-1}(P_j))}{P_j}\right)$ is strictly positive (Lee, 2007).

Women are positively selected in both informal and formal jobs.[13] As for men, they are positively selected in informal employment but negatively selected in formal employment according to both Lee's and DMF's methods.[14] This is consistent with the conclusions of papers finding that some workers have an individual comparative advantage in informal jobs and would not have higher earnings in formal jobs (Maloney, 1999, 2004 among others). For given values of observable characteristics, men holding informal jobs have unobserved characteristics that are on average most valued in this sector. Consequently, observed wages overestimate male wage offers in informal jobs. By contrast, men are on average negatively selected into the formal sector. Those with the highest wage potentials in formal jobs do not self-select into those jobs and choose other work statuses (note that I do not consider self-employed workers

[13] In Table 7, Column (3) shows that $\sigma^2\rho_2 < 0$ which implies a positive selection bias in informal jobs according to Lee's method. In column (5), given that DMF's approach imposes the restriction $\sum_k \rho_k = 0$, $\rho_2 > 0$ and $\sigma^2\rho_2 > 0$ which also corresponds to a positive bias in informal jobs. The same reasoning applies for the female selection rule in formal jobs.

[14] Focusing on the male selection rule into formal jobs, column (4) shows that $\sigma^2\rho_3 > 0$ which implies a negative bias according to Lee's method. In column (6), given that of Table 8 DMF's approach imposes the restriction $\sum_k \rho_{jk} = 0$, $\rho_3 < 0$ and $\sigma^2\rho_3 < 0$ which also corresponds to a negative selection bias in formal jobs.

Table 7
Female hourly wages in the informal and formal sectors

	OLS		Selection Lee		Selection DMF	
	Informal (1)	Formal (2)	Informal (3)	Formal (4)	Informal (5)	Formal (6)
Years of education	0.033**	0.067**	0.020**	0.078**	0.010*	0.093**
	(0.002)	(0.002)	(0.003)	(0.002)	(0.004)	(0.004)
Age	0.032**	0.020**	0.042**	0.029**	0.037**	0.029**
	(0.004)	(0.003)	(0.004)	(0.002)	(0.006)	(0.002)
Age^2	−0.000**	−0.000**	−0.000**	−0.000**	−0.000**	−0.000**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Firm-specific experience (in years)	−0.010	0.008**	−0.009	0.008**	−0.009	0.008*
	(0.006)	(0.003)	(0.006)	(0.003)	(0.006)	(0.003)
Firm-specific experience^2	0.001	−0.001**	0.001	−0.001**	0.001	−0.001**
	(0.001)	(0.000)	(0.001)	(0.000)	(0.001)	(0.000)
Black	−0.040	−0.088**	−0.019	−0.088**	−0.015	−0.099**
	(0.020)	(0.009)	(0.016)	(0.007)	(0.015)	(0.007)
Unemployment rate	−3.907**	−6.800**	−3.177**	−7.038**	−2.665**	−6.949**
(regional, by education group)	(0.479)	(0.373)	(0.395)	(0.144)	(0.495)	(0.144)
Constant	0.673**	1.333**	0.077	0.939**	0.226	0.735**
	(0.079)	(0.119)	(0.102)	(0.085)	(0.151)	(0.083)
σ^2			0.664**	0.271**	0.364**	0.362**
			(0.086)	(0.009)	(0.016)	(0.071)
ρ_1					0.042	0.346**
					(0.082)	(0.088)
ρ_2			−0.340**			−0.790**
			(0.024)			(0.103)
ρ_3				−0.277**	−0.398**	
				(0.034)	(0.075)	
R^2	0.21	0.47				
N	8,955	30,217	8,955	30,217	8,955	30,217

Notes: All regressions include sector and region dummies. * $p < 0.05$; ** $p < 0.01$. Standard errors in parenthesis. Clustered s.e. in OLS regressions. Bootstrap estimates of the s.e. when controlling for selection to account for the two-step procedure. In Lee's approach, a negative $\sigma_j \rho_j$ implies a positive selection bias as $\sigma_j \rho_j h(P_j)$ is then strictly positive. The DMF approach imposes the restriction $\sum_k \rho_{jk} = 0$ so that ρ_2 in column (5) and ρ_3 in column (6) are both positive. The selection bias is positive in both formal and informal jobs using both approaches.

Table 8
Male hourly wages in the informal and formal sectors

	OLS		Selection Lee		Selection DMF	
	Informal (1)	Formal (2)	Informal (3)	Formal (4)	Informal (5)	Formal (6)
Years of education	0.046**	0.072**	0.029**	0.059**	0.013*	0.069**
	(0.002)	(0.002)	(0.005)	(0.002)	(0.005)	(0.002)
Age	0.033**	0.036**	0.027**	0.016**	−0.002	0.016**
	(0.004)	(0.004)	(0.001)	(0.000)	(0.004)	(0.001)
Age^2	−0.000**	−0.000**	−0.000**	−0.000**	−0.000	−0.000**
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Firm-specific experience (in years)	−0.015*	0.008**	−0.013**	0.008**	−0.014	0.007**
	(0.006)	(0.003)	(0.001)	(0.001)	(0.008)	(0.001)
Firm-specific experience^2	0.002*	−0.001**	0.002**	−0.001**	0.002*	−0.001**
	(0.001)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)
Black	−0.060**	−0.098**	−0.058**	−0.101**	−0.060**	−0.100**
	(0.017)	(0.010)	(0.010)	(0.002)	(0.021)	(0.001)
Unemployment rate	−3.320**	−6.578**	−3.182**	−6.881**	−3.273**	−6.627**
(regional, by education group)	(0.526)	(0.378)	(0.716)	(0.139)	(0.418)	(0.228)
Constant	0.800**	1.130**	0.641**	1.867**	1.343**	1.575**
	(0.084)	(0.077)	(0.127)	(0.021)	(0.116)	(0.065)
σ^2			0.683**	0.270**	0.571**	0.507**
			(0.132)	(0.000)	(0.048)	(0.004)
ρ_1					0.496**	0.797**
					(0.063)	(0.002)
ρ_2			−0.393**			−0.718**
			(0.017)			(0.015)
ρ_3				0.546**	−0.780**	
				(0.037)	(0.030)	
R^2	0.28	0.44				
N	8,366	35,591	8,366	35,591	8,366	35,591

Notes: All regressions include sector and region dummies. * $p < 0.05$; ** $p < 0.01$. Standard errors in parenthesis. Clustered s.e. in OLS regressions. Bootstrap estimates of the s.e. when controlling for selection to account for the two-step procedure. In Lee's approach, a negative $\sigma_j \rho_j$ implies a positive selection bias as $\sigma_j \rho_j h(P_j)$ is then strictly positive. The DMF approach imposes the restriction $\sum_k \rho_{jk} = 0$ so that ρ_2 in column (5) is positive and ρ_3 in column (6) is negative. The selection bias is positive in informal jobs and negative in formal jobs using both approaches.

and employers). Negative selection occurs when the reservation wage increases with the wage offer.

The selection pattern of men remains the same across different education levels. For women however, the average selection rule hides heterogeneity across education groups.[15] The positive selection bias in formal jobs holds for women with secondary education or less but not for women with tertiary education, who are negatively selected into formal jobs as men are. Highly educated women working in the formal sector have lower wage potential compared to highly educated women in other work statuses. The implications of the different selection patterns across gender for the estimated gender wage gaps in formal and in informal jobs are discussed in the next section.

(d) The gender wage gap in informal and formal jobs

Table 9 displays the estimated gender wage gaps in formal and informal jobs for the whole population as well as for different education groups. Figures in bold indicate that the difference between the formal and the informal wage gaps is significant at 10% according to Welch's test.

Panel 1 shows that the raw (or total) gender wage gaps are positive and significant in both formal and informal jobs. The average raw gap is 5% among informal employees and not significantly different from the raw wage gap of 8% among formal employees.[16] However, the raw wage gaps conceals different compositions of the male and female labor force. When I estimate the gaps for different education groups, which already account for some differences in productive characteristics, I see that the gender wage gap in informal jobs is smaller than the gap in formal jobs, in particular for workers with primary education or no education. A bigger gender wage gap among employees with a legal contract may seem at odds with the intuition that labor market regulation, in particular a minimum wage, should reduce the scope for wage gaps at the bottom of the wage distribution. Note, however that the minimum wage also serves as a reference for employers determining informal employees' wages. This so-called "lighthouse" effect implies that the minimum wage is actually most binding in informal jobs where the lowest wages are observed. Another striking pattern is the increase in the gender wage gap with the education level, which could be interpreted as a form of "glass ceiling" in both formal and informal jobs. That being said, the raw wage gap does not properly account for the heterogeneity of the labor force. As men and women might have different characteristics in both types of jobs, the next panels provide a more detailed analysis.

Panel 2 of Table 9 shows that controlling for observable characteristics such as the exact number of years of education, age, tenure within the firm, sector of activity and location, increases the wage gap in both formal and informal jobs (from panel 1 to panel 2). This result is expected as women are more educated than men and women in the labor force present overall better observable characteristics on average than men in this sample. The adjusted wage gaps in formal and informal jobs are not statically different from one another. Although skills receive lower returns in informal jobs, the gender difference in returns is about the same in percentage terms in formal and informal jobs: the average wage gap is about 0.2 log points which amounts to a difference of 22%. If I look at the gaps for different education groups, adjusting women's returns to the returns obtained by men would increase women's wages by 19% (11%) among formal (informal) female employees with primary education. This would increase wages by 22% (24%) among formal (informal) female employees with secondary education. For high-skilled workers, the gap is significantly larger among formal employees at 30% compared to 19% among informal employees. The so-called "glass ceiling effect" is stronger in jobs with registered labor contracts which points in the direction of the hypothesis formulated in Section 2.

I now turn to the effect of the selection bias. The data give information on observed wages for workers in a given type of job. To infer the magnitude of the wage gap correctly though, I want to compare wage *offers* (that would be) made to all men and women. If selection into formal *vs.* informal jobs is non-random, observed wages either overstate or understate wage offers. If the selection bias differs by gender, the observed raw wage gap, given in the first panel of Table 9, will not reflect the raw difference in wage offers. Controlling for sorting across work statuses enables me to recover the average wage offers within each type of jobs, providing the control function captures the selection bias properly. Panel 3 shows how selection leads to different changes in the average gender wage gap in informal and formal jobs.

In the informal sector, the observed wage gap overstates the wage gap in wage *offers*. The observed wage gap among informal employees is 0.049 log points in panel 1 but in panel 3.1 the gap in wage offers is 0.014 (−0.024) log points according to Lee's (DMF's) method and is not significantly different from zero. The results on the wage regressions in Tables 7 and 8 showed that both men and women are positively selected into informal jobs. This implies that observed informal wages overestimate informal wage offers for both men and women; but more so for men. Indeed, the term $\sigma^2\rho_2$ is greater in magnitude for men (columns (3) and (5) in Table 8) than for women (columns (3) and (5) in Table 7). As a result, controlling for selection into informal jobs reduces the average informal wage offer more for men than for women, and thus reduces the wage gap (panel 3.1). Among informal employees, differences in returns have no role in explaining the gap anymore (panel 3.2). The adjusted gap in wage offers is 0.144 (0.100) log points according to Lee's (DMF's) method but it is not statistically significant. In other words, after purging the estimates from the selection effect, I cannot reject the hypothesis that men and women receive equal compensation for their skills in informal jobs.

In the formal sector, on the other hand, the gender gap in wage offers is offset by the selection bias and is underestimated by the observed gender wage gap. While the observed gender wage gap among formal employees is 0.73 log points, the gender gap in wage offers is 0.333 (0.289) log points according to Lee's (DMF's) method which corresponds to an increase in the gender gap from 8% to about 35% (panel 1 to panel 3.1). This is because men tend to be negatively selected into formal jobs (Table 8) meaning that observed male wages underestimate male wage offers while women are positively selected into formal jobs (Table 7) meaning that observed female wages overestimate female wage offers. Observable characteristics are better among women, which makes the share of the gender wage gap resulting from different returns even bigger than the total wage difference adjusted for the selection bias (panel 3.2). The gender gap in wage offers due to different returns in characteristics is 0.475 log points (60%) according to Lee's method and 0.441 log points (55%) according to DMF's method. Moreover, the increase in the wage gap with higher education levels is robust to the treatment of selection. For high-skilled wage-earners, however, the adjusted wage gaps are not statistically different in the two types of jobs. This is due to the very small sample size of informal wage-earners with tertiary education and thus the imprecise estimate of the gap for this group.

These results are robust to different exclusion restrictions in the treatment of selection. Table 10 gives the gender wage gap

[15] Results available upon request.

[16] Results in Table 9 are expressed on the logarithmic scale. To obtain the difference in percentage points: $(\exp(WG) - 1) \times 100$.

Table 9
Gender wage gap decomposition. Informal and formal sectors, urban areas

Level of Education		All	Primary or less	Secondary	Tertiary
1-Total wage gap: $\overline{TWG}_j = \overline{\ln W}_{mj} - \overline{\ln W}_{fj}$					
	Informal	0.049**	**−0.069****	**0.106****	0.227**
		(0.015)	(0.024)	(0.016)	(0.052)
	Formal	0.073**	**0.177****	**0.201****	0.246**
		(0.017)	(0.019)	(0.011)	(0.013)
	Welch's t-statistics	1.03	8.48	5.54	0.66
2-Controlling for observables only					
Part due to differences in returns: $\overline{WG}_j = \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}}_{pj}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}}_{pj} - \widehat{\boldsymbol{\beta}}_{fj})$					
	Informal	0.188**	**0.103****	0.216**	**0.176****
		(0.020)	(0.033)	(0.017)	(0.029)
	Formal	0.217**	**0.171****	0.199**	**0.261****
		(0.011)	(0.017)	(0.010)	(0.016)
	Welch's t-statistics	0.17	2.34	−1.02	2.60
3-Gap in wage offers. Controlling for observables and self-selection					
3.1-Total gap in wage offers: $\overline{TWG}_{Sj} = \overline{\ln W}_{mj} - \overline{\ln W}_{fj} - (\theta_{mj}h_{mj} - \theta_{fj}h_{fj})$					
Lee	Informal	**0.014**	0.057	**0.009**	−0.683
		(0.109)	(0.132)	(0.134)	(0.807)
	Formal	**0.333****	0.128*	**0.383****	0.577**
		(0.020)	(0.059)	(0.022)	(0.050)
	Welch's t-statistics	2.87	0.49	2.75	1.56
DMF	Informal	**−0.024**	0.018	**−0.017**	−0.435
		(0.079)	(0.125)	(0.095)	(0.669)
	Formal	**0.289****	0.089	**0.347****	0.349**
		(0.018)	(0.077)	(0.022)	(0.059)
	Welch's t-statistics	3.85	0.49	3.73	1.17
3.2-Part due to difference in returns: $\overline{WG}_{Sj} = \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}_{pj}}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}_{pj}} - \widehat{\boldsymbol{\beta}_{fj}})$					
Lee	Informal	**0.144**	0.228	0.112	−0.737
		(0.108)	(0.131)	(0.135)	(0.807)
	Formal	**0.475****	0.120*	0.376**	0.587**
		(0.020)	(0.058)	(0.022)	(0.051)
	Welch's t-statistics	3.02	−0.76	1.93	1.64
DMF	Informal	**0.100**	0.171	**0.080**	−0.490
		(0.079)	(0.125)	(0.095)	(0.672)
	Formal	**0.441****	0.080	**0.342****	0.359**
		(0.017)	(0.078)	(0.022)	(0.060)
	Welch's t-statistics	4.20	−0.62	2.67	1.26
Number of informal workers		17,321	5,574	10,289	1,458
Share of women		52%	49%	53%	55%
Number of formal workers		65,808	8,626	39,311	17,871
Share of women		46%	34%	42%	60%

Notes: * $p < 0.05$; ** $p < 0.01$. s.e. in parenthesis. Panel 1: Eqn. (1). Panel 2: Eqn. (3). Panel 3.2: Eqn. (7).The results are expressed on the logarithmic scale. To obtain the difference in percentage points: $(\exp(WG) - 1) \times 100$. Figures in bold indicate that the difference between the formal and informal wage gap is significant at 10% when $|t| > 1.64$, the difference is significant at 5% if $|t| > 1.96$.

adjusted for selection on unobservables without using children as an excluded variable. Gender differences in wage offers among informal wage-earners remain not statically different from zero. This result holds using two different approaches controlling for selection and across all levels of education. Among formal wage-earners however, both the total difference in wage offers and the gap due to different returns on characteristics are positive and significant. I also find here that the gender wage gaps are higher among formal employees with secondary education or higher compared to formal employees with less than secondary education.

These results highlight the fact that labor regulations may impact gender wage inequality in the urban labor market in Brazil. The finding that wage gaps are positive and significant only in formal jobs is consistent with the following explanation. If employers believe that women have a higher probability of quitting, statistical discrimination induces employers to pay women lower wages because they expect higher average female labor costs. I argue that the gap in expected labor costs because of gender differences in labor market attachment is higher in jobs where employment protection is binding. When an employee goes on temporary leave, his/her job must remain available to him/her, generating costs due to vacancy or temporary replacement. This effect is expected to be weaker in informal jobs because the job of the employee on leave can be allocated to another worker. I also find that the wage gap is higher among high-skilled workers in formal jobs, a result that is commonly found in the literature on gender wage gaps. This finding is often explained by statistical discrimination that produces higher gender gaps in high-wage jobs. Higher gender differences in pay among formal workers and the increase in the pay gap with the education level are thus consistent.

(e) Self-employment

So far I have focused on wage earners to compare formal and informal workers with the same employment status. However, self-employment can also be considered as part of the informal labor market (see Henley *et al.,* 2009 on the differences in the definitions of informality in the case of Brazil). To provide a larger picture of gender inequality in earnings, I also consider self-employed individuals and reports the results in

Appendix C. I treat self-employment as a different category and estimate separate earnings equations for female and male self-employed workers (Table 11). Then I compute the gender gap in earnings for this group (Table 12). The raw earning gap between self-employed females and males is 13% (0.12 log points), higher than the raw wage gap among employees. The raw gap in earnings is similar across education groups which is in contrast with the pattern observed among informal employees for whom the raw gender wage gap increases with education. Controlling for a full set of observable characteristics, the adjusted gap in earnings among self-employed workers increases to 22%, which is similar to the gender wage gap among employees. The adjusted gender gap is higher among self-employed workers with secondary education or higher compared to self-employed workers with lower educational attainments, a pattern that is similar to what we observe among employees.

Panel 3 shows the gender gaps in *potential* earnings controlling for selection into employment status. The selection model now includes four potential outcomes instead of three (not in paid employment, formal and informal wage employment, and self-employment). In this case, Lee's and DMF's methods lead to different results. Lee's (1983) approach assumes that the selection bias in the earnings equation of self-employed is well controlled for by only using the probability of being self-employed. Dubin and McFadden (1984) make less restrictive assumptions and allow the selection bias to depend on the probability of having another employment status, here formal and informal employment or not in paid employment.[17] While Lee's assumption seems to be reasonable for informal and formal employees, it does not hold in the case of self-employment and misses part of the selection bias in the self-employed earnings equation. This can be seen in the results of the estimation of the earnings equation using DMF's method in Table 11 in the Appendix C. The correlation of the error term of the earning equation with the error terms in the selection equation for the three alternative employment statuses are significant and change the selection rule for self-employment.

The gender gap in potential earnings remains positive and significant according to Lee's method but it is not significantly different from zero when controlling for selection with the more flexible method (panel 3.1). Looking at the gap in potential earnings due to different returns to characteristics (panel 3.2), the average gap and the gap among those with primary education or less are high and significant according to Lee's method but again not significantly different from zero according to DMF's method. Overall, I find that the total gender gap in earnings as well as the gender gap adjusted for observable characteristics are also positive among the self-employed but that, after having adjusted for selection on unobservables, potential earnings are similar for self-employed men and self-employed women. In this respect, gender inequality in earnings among the self-employed resembles gender wage inequality among informal employees.

5. Conclusion

This paper investigates gender wage inequality in formal and informal jobs in Brazil. The data show that the average raw gender wage gap is positive and significant in both sectors. The informal sector features the highest total average gender wage gap but this conceals differences in male and female characteristics. When I ignore the selection bias, the differences in returns to productive characteristics are the same in formal and informal jobs and are responsible for about 22% of the pay gap. The paper additionally shows that the similarity between the formal and informal wage gap is artificially generated by the different selection of men and women into formal and informal jobs. In the informal sector, observed average wages for both men and women overestimate their respective average wage offers, though more so for men. The difference in average wage offers faced by men and women can in fact be entirely explained by differences in selection bias. As a result, the gender gap due to different returns on characteristics is not significant in the informal sector. The opposite happens in the formal sector. The gender difference in selection bias narrows the gender gap in observed wages. This is because women are positively selected into formal jobs but men are not. As a result, observed female wages are higher that the wage offers that would be made to all women. After controlling for selection, the gender wage gap due to different returns on characteristics is even bigger in formal jobs. Moreover, the gender wage gap increases with the level of education in the formal sector.

Bigger gender differences in returns on productive characteristics in the formal sector should certainly not lead to the conclusion that employment protection legislation is detrimental to women. First, formal jobs provide higher wages for women, even if the formal wage premium is lower for women than for men. Additionally, given that women face a higher unemployment rate and need to take maternity leave, the flow of earnings of women relative to men can be higher in the formal sector because unemployment benefits and maternity leave benefits compensate for wage losses while this does not happen in the informal sector. Further research is needed, first to identify the impact of labor regulation on discriminatory behavior and, second, to investigate how participating in the informal sector affects gender differences in earnings over the life cycle.

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Appendix A. Empirical model

First, I further assume that the errors of the selection equation are independent and identically distributed following a type I extreme value distribution, the probability of being in status j for individual i is defined by the multinomial logit model (McFadden, 1973) as in Eqn. (8).

$$P_{ij} = Pr(Y_i = j) = \frac{\exp(\mathbf{X}_{1i}\boldsymbol{\lambda}_j + \mathbf{Z}_i\boldsymbol{\alpha}_j)}{\sum_j^N \exp(\mathbf{Z}_i\alpha_j)}. \tag{8}$$

The independence from irrelevant alternatives is a strong assumption of the logit model. It implies that the odds ratio between two alternatives are independent of other alternatives. In the case under investigation here, as an illustration, it implies that if all women not in the labor force were to enter the labor

[17] More precisely, Lee (1983) assumes that the correlation between the error term in the self-employed earnings' equation u_{se} and the error terms in the selection equation $(\mu_j - \mu_{se})$ is of the same sign for all$j = \{\text{not in paid employment, informal, formal}\}$. In other words, unobservable determinants of the choice of self-employment against any other employment status should be correlated in the same direction with unobservable determinants of self-employment earnings. Dubin and McFadden (1984) do not impose this restriction and the selection bias may depend on the correlation of the error term of the self-employed earnings equation u_{se} with the error terms of each of the four selection equations.

market, the repartition of women in formal and informal jobs would remain unchanged. Although not completely implausible, this assumption is clearly restrictive. Bourguignon *et al.* (2007) evaluate selection bias correction techniques based on the multinomial logit model and show with Monte Carlo experiments that those methods provide good bias correction in the outcome equation even when the IIA hypothesis is violated.

Adopting Lee's (1983) approach, I first assume that the joint distribution of u_j and a transformation[18] of μ_j does not depend on the other μ_k for $k \neq j$. Under this assumption and an additional linearity assumption the expected value of u_j, conditional on category j being observed is: [19]

$$E(u_j|Y = j, \mathbf{X}_1, \mathbf{Z}) = \sigma_j \rho_j \left(-\frac{\phi(\Phi^{-1}(P_j))}{P_j} \right),$$

with ϕ and Φ the standard normal density and cumulative distribution functions, σ_j is the standard deviation of the wage errors, and ρ_j is the correlation coefficient between the errors of the outcome equation and the errors of the wage equation in sector j. The control function is $h = -\frac{\phi(\Phi^{-1}(P_j))}{P_j}$ and $\sigma_j \rho_j$ is estimated by least squares. Only one correlation parameter ρ_j is estimated per wage equation under this method.

As a second approach to dealing with selection, I also follow Dubin and McFadden (1984) who make less restrictive assumptions on the correlation between u_j and the $(\mu_k - \mu_j)$. In particular, these allow the selection bias to originate in the correlation of u_j not only with μ_j but also with μ_k for $k \neq j$. The linearity assumption on the conditional mean of the wage equation residuals is as follows:

$$E(u_j|Y = j, \mathbf{X}_1, \mathbf{Z}) = \sigma_j \frac{\sqrt{6}}{\pi} \sum_k \rho_{jk} (\mu_k - E(\mu_k)),$$

where j is the final outcome and $k = 1, \ldots 3$ all the potential outcomes. ρ_{jk} is the correlation coefficient between u_j and μ_k and Dubin and McFadden (1984) make the restriction that the correlation coefficients sum up to zero $\sum_k \rho_{jk} = 0$. Given the multinomial logit formulas it follows that[20]:

$$E(\mu_j - E(\mu_j)|V_j > \max_{s \neq j}(V_s), Z) = -\ln(P_j),$$

$$E(\mu_k - E(\mu_k)|V_j > \max_{s \neq j}(V_s), Z) = \frac{P_k \ln(P_k)}{1 - P_k}, \quad \text{for } k \neq j.$$

Appendix B. Robustness check

Table 10
Decomposition of the gender gap in wage offers among informal and formal employees in urban areas. Alternative exclusion restrictions

Level of Education		All	Primary or less	Secondary	Tertiary
3-Gap in wage offers. Controlling for observables and self-selection					
3.1-Total gap in wage offers: $\overline{TWG}_{Sj} = \overline{\ln W}_{mj} - \overline{\ln W}_{fj} - (\theta_{mj} h_{mj} - \theta_{fj} h_{fj})$					
Lee	Informal	−**0.094**	0.019	−**0.113**	−**1.349**
		(0.113)	(0.160)	(0.136)	(0.822)
	Formal	**0.293****	0.136*	**0.378****	**0.488****
		(0.028)	(0.063)	(0.030)	(0.059)
	Welch's *t*-statistics	3.33	0.68	3.53	2.23
DMF	Informal	−**0.067**	−0.021	−**0.053**	−0.613
		(0.080)	(0.126)	(0.091)	(0.710)
	Formal	**0.263****	0.105	**0.345****	0.322**
		(0.019)	(0.066)	(0.030)	(0.059)
	Welch's *t*-statistics	4.03	0.89	4.14	1.31
3.2-Part due to difference in returns: $\overline{WG}_{Sj} = \overline{\mathbf{X}}'_m(\widehat{\beta}_{mj} - \widehat{\beta}_{pj}) + \overline{\mathbf{X}}'_f(\widehat{\beta}_{pj} - \widehat{\beta}_{fj})$					
Lee	Informal	**0.037**	0.191	−**0.010**	−**1.404**
		(0.112)	(0.157)	(0.136)	(0.818)
	Formal	**0.435****	0.128*	**0.371****	**0.497****
		(0.028)	(0.062)	(0.031)	(0.061)
	Welch's *t*-statistics	3.44	−0.37	2.73	2.32
DMF	Informal	**0.056**	0.131	**0.043**	−0.668
		(0.081)	(0.125)	(0.089)	(0.713)
	Formal	**0.414****	0.097	**0.339****	0.331**
		(0.019)	(0.067)	(0.030)	(0.058)
	Welch's *t*-statistics	4.29	−0.24	3.15	1.40
Number of informal workers		17,253	5,551	10,251	1,451
Share of women		52%	49%	53%	55%
Number of formal workers		65,586	8,589	39,189	17,817
Share of women		46%	34%	42%	60%

Notes: This table presents a sensitivity check of the benchmark results presented in Panel 3 of Table 9. The control function is constructed without children in the set of excluded variables. Panel 3.2: Eqn. (7). * $p < 0.05$; ** $p < 0.01$. s.e. in parenthesis. The results are expressed on the logarithmic scale. To obtain the difference in percentage points: $(\exp(WG) - 1) \times 100$. Welch's test is applied to test the difference between the formal and the informal gaps with different population sizes and variances. Figures in bold indicate that the difference between the formal and the informal wage gaps is significant at 10% when $|t| > 1.64$, the difference is significant at 5% if $|t| > 1.96$.

[18] The transform is defined as $\Phi^{-1}(F(\mu_j|\mathbf{X}_1, \mathbf{Z}))$ with Φ the standard normal cumulative distribution function and F the cumulative distribution function of μ_j.

[19] See Lee (1983) for a proof.

[20] See Dubin and McFadden (1984) for a proof.

Appendix C. Self-employment

Table 11
Hourly earnings for female and male self-employed workers

	OLS		Lee		DMF	
	Female (1)	Male (2)	Female (3)	Male (4)	Female (5)	Male (6)
Years of education	0.055**	0.053**	0.057**	0.053**	0.053**	0.024**
	(0.004)	(0.001)	(0.001)	(0.002)	(0.009)	(0.003)
Age	0.032**	0.037**	0.042**	0.038**	0.045**	0.012*
	(0.005)	(0.003)	(0.009)	(0.006)	(0.008)	(0.005)
Age^2	−0.000**	−0.000**	−0.000**	−0.000**	−0.000**	−0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Tenure (in years)	−0.009	0.015*	−0.009	0.015*	−0.008	0.015**
	(0.009)	(0.007)	(0.007)	(0.007)	(0.008)	(0.005)
Tenure^2	0.000	−0.002**	0.000	−0.002	0.000	−0.002**
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Black	−0.117**	−0.125**	−0.122**	−0.125**	−0.108**	−0.144**
	(0.016)	(0.010)	(0.016)	(0.016)	(0.014)	(0.014)
Unemployment rate	−3.754**	−2.561**	−3.469**	−2.572**	−3.681**	−2.852**
(regional, by education group)	(0.446)	(0.354)	(0.447)	(0.104)	(0.431)	(0.328)
Constant	1.071**	0.685**	0.520	0.656**	0.633**	1.067**
	(0.083)	(0.078)	(0.336)	(0.037)	(0.229)	(0.151)
σ^2			0.640**	0.413**	0.642**	1.048**
			(0.140)	(0.010)	(0.130)	(0.120)
ρ_1					−0.410**	0.657**
					(0.097)	(0.102)
ρ_2					0.532*	0.074
					(0.247)	(0.177)
ρ_3					−0.167	−0.865**
					(0.248)	(0.045)
ρ_4			−0.214*	−0.018*		
			(0.088)	(0.007)		
R^2	0.27	0.29				
N	8,812	16,564				

Notes: All regressions include sector and region dummies. * $p < 0.05$; ** $p < 0.01$. Standard errors in parenthesis. Clustered s.e. in OLS regressions. Bootstrap estimates of the s. e. when controlling for selection to account for the two-step procedure. In Lee's approach, a negative $\sigma_j\rho_j$ implies a positive selection bias as $\sigma_j\rho_j h(P_j)$ is then strictly positive. The DMF approach imposes the restriction $\sum_k \rho_{jk} = 0$ so that ρ_4 are negative for both men and women. The correlation coefficient between the error terms of the earnings equation and the selection equation is positive using Lee's approach but negative using DMF's approach.

Table 12
Decomposition of the gender gap in earnings among the self-employed in urban areas

Level of Education		All	Primary or less	Secondary	Tertiary
1-Total wage gap:					
$\overline{TWG}_j = \overline{\ln W}_{mj} - \overline{\ln W}_{fj}$		**0.120****	0.197**	0.211**	0.195**
		(0.019)	(0.028)	(0.021)	(0.044)
Welch's *t*-stat	Formal *vs.* Self-employed	−1.86	−0.66	−0.52	1.74
	Informal *vs.* Self-employed	2.94	7.19	4.00	−0.47
2-Controlling for observables only					
Part due to differences in returns:					
$\overline{WG}_j = \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}}_{pj}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}}_{pj} - \widehat{\boldsymbol{\beta}}_{fj})$		0.208**	0.146**	0.219**	0.217**
		(0.019)	(0.030)	(0.022)	(0.035)
Welch's *t*-stat	Formal *vs.* Self-employed	0.05	0.80	−1.93	1.48
	Informal *vs.* Self-employed	0.10	0.17	0.01	0.14
3-Gap in wage offers. Controlling for observables and self-selection					
3.1-Total gap in wage offers: $\overline{TWG}_{Sj} = \overline{\ln W}_{mj} - \overline{\ln W}_{fj} - (\theta_{mj}h_{mj} - \theta_{fj}h_{fj})$					
Lee's approach		**0.119****	**0.197****	0.211**	0.195**
		(0.009)	(0.018)	(0.012)	(0.027)
Welch's *t*-stat	Formal *vs.* Self-employed	−2.11	−1.87	−0.46	0.65
	Informal *vs.* Self-employed	3.22	2.47	1.72	1.44
DMF's approach		−0.114	−0.069	−0.083	−0.104
		(0.134)	(0.312)	(0.198)	(0.360)
Welch's *t*-stat	Formal *vs.* Self-employed	4.20	0.12	3.13	1.11
	Informal *vs.* Self-employed	−0.77	−0.92	−0.70	−0.88
3.2-Part due to difference in returns: $\overline{WG}_{Sj} = \overline{\mathbf{X}}'_m(\widehat{\boldsymbol{\beta}}_{mj} - \widehat{\boldsymbol{\beta}}_{pj}) + \overline{\mathbf{X}}'_f(\widehat{\boldsymbol{\beta}_{pj}} - \widehat{\boldsymbol{\beta}_{fj}})$					
Lee's approach		0.477**	0.723*	0.279	0.011
		(0.135)	(0.319)	(0.196)	(0.418)

Table 12 (*continued*)

Level of Education		All	Primary or less	Secondary	Tertiary
Welch's t-stat	Formal *vs.* Self-employed	−1.71	−1.67	−0.51	0.64
	Informal *vs.* Self-employed	2.94	1.89	1.35	1.50
DMF's approach		−0.056	−0.138	-0.094	-0.083
		(0.134)	(0.314)	(0.200)	(0.352)
Welch's *t*-stat	Formal *vs.* Self-employed	4.79	0.27	3.15	1.09
	Informal *vs.* Self-employed	−1.12	−1.46	−1.12	−0.83
Number of self-employed		25,376	7,050	14,530	3,762
Share of women		36%	24%	37%	45%

Notes: Panel 1: Eqn. (1). Panel 2: Eqn. (3). Panel 3.2: Eqn. (7). Gaps in bold are statistically different at 5% from the gaps in the two other categories according to Welch's test.

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